Board 4: Instrument Droid

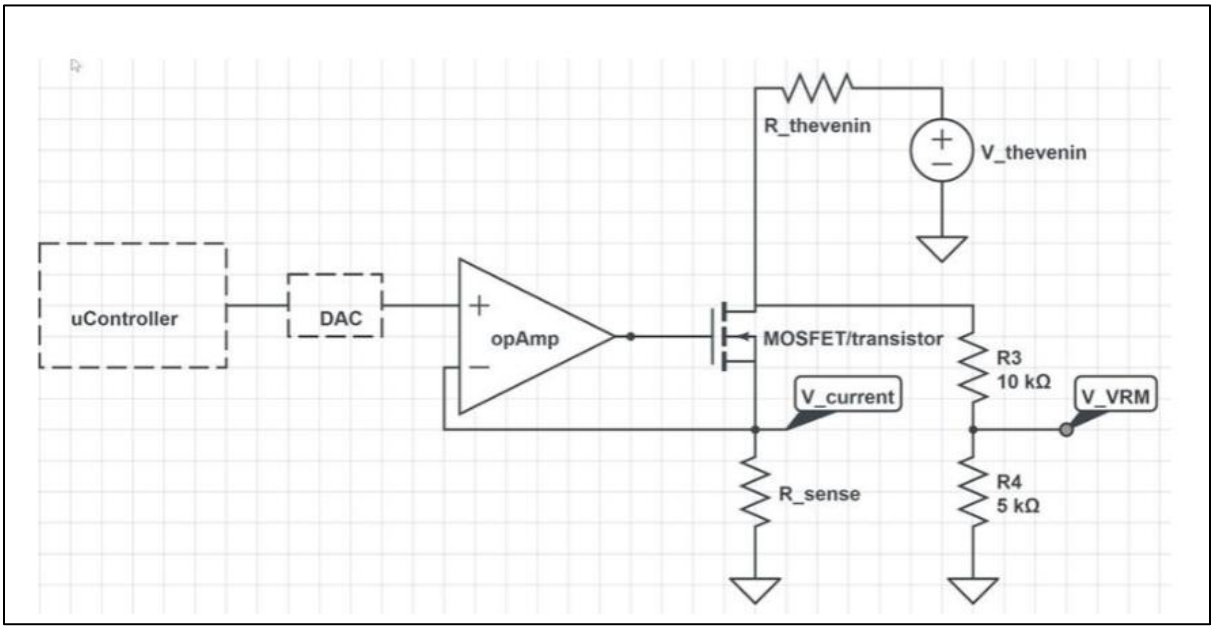
Objective:

To design and implement a 4-layer standalone board using the ATmega328 microcontroller that can measure the Thevenin equivalent resistance of various DC voltage sources by sensing voltage and current under controlled load conditions.

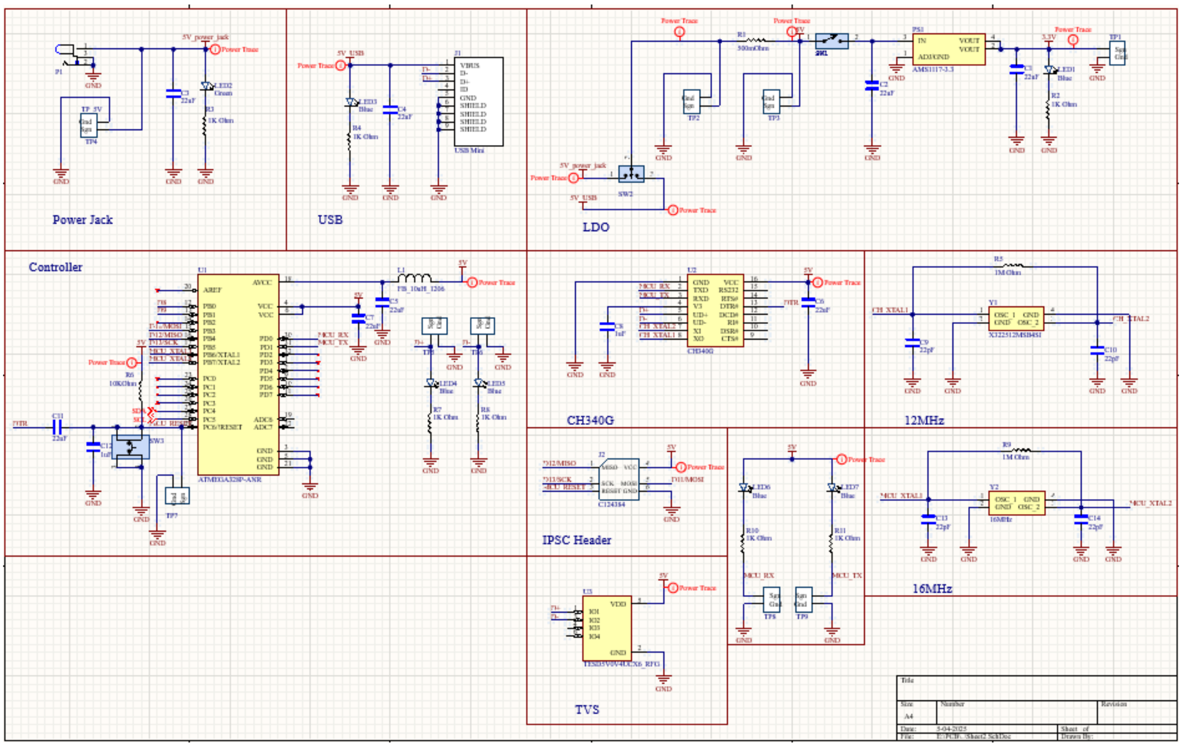
Bill of Material:

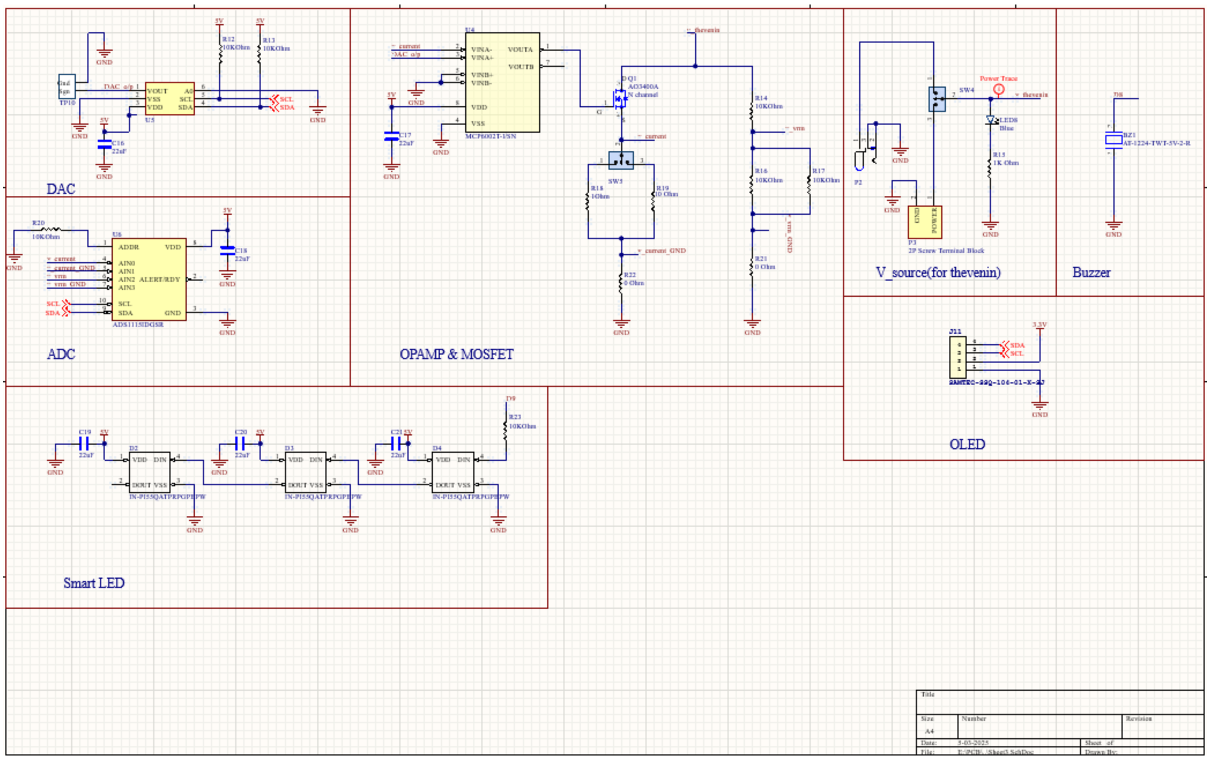
1	Comment	Description	Designator	Quantity
2	AT-1224-TWT-5V-2-R	Buzzers Transducer, Externally Driven Electromechanical/Magnetic 5 V 40mA 2.4kHz 87dB @ 5V, 10cm Through Hole PC Pins	BZ1	1
3	22uF	22uF ±10% 25V X5R 1206 Multilayer Ceramic Capacitors MLCC - SMD/SMT RoHS	C1, C2, C3, C4, C5, C6, C7, C11 , C16, C17, C18, C19, C20, C21	14
4	1uF	MULTILAYER CERAMIC CAPACITORS MLCC - SMD/SMT 1UF 50V 1206 ROHS	C8, C12	2
5	22pF	22pF ±5% 1kV C0G 1206 Multilayer Ceramic Capacitors MLCC - SMD/SMT RoHS	C9, C10, C13, C14	4
6	IN-PI55QATPRPGPBP	LIGHT EMITTING DIODES (LED) RGB 620~630NM RED, 520~530NM GREEN, 465~475NM BLUE 600MCD@16MARED,1200MCD@16MAGRI	D2, D3, D4	3
7	USB Mini	Through Hole USB Connectors RoHS	J1	1
8	NRPN032PAEN-RC	Connector Header Through Hole 6 position 0.079" (2.00mm)	J2	1
9	SAMTEC-SSQ-104-01-X	2.54mm 3A bronze -40℃~+105℃ Straight 1x4P Top 4 1 Square Holes Straight,P=2.54mm Female Headers ROHS	J11	1
10	FB_10uH_1206	25mA 10uH ±10% 800mΩ 1206 Inductors (SMD) ROHS	L1	1
11	Blue	blue 1206 Light Emitting Diodes (LED) ROHS	LED1, LED3, LED4, LED5, LED6, LED7, LED8	7
12	Green	Green 513~528nm 1206 Light Emitting Diodes (LED) RoHS	LED2	1
13	Power Jack	Power Barrel Connector Jack 2.10mm ID (0.083"), 5.50mm OD (0.217") Through Hole, Right Angle	P1, P2	2
14	2P Screw Terminal Blo	Green Straight pin 3.5mm pitch; P=3.5mm Screw terminal ROHS	P3	1
15	AMS1117-3.3	LOW DROPOUT REGULATORS(LDO) POSITIVE FIXED 1.3V @ 800MA 15V 3.3V 1A SOT-223 ROHS	PS1	1
16	AO3400A	MOSFET N Trench 30V 5.7A 1.5V @ 250uA 26.5 m? @ 5.7A,10V SOT-23-3L RoHS	Q1	1
17	500m	Resistors/Shunt 250mW 500mΩ ±800ppm/℃ ±5% 1206 Current Sense Resistors ROHS	R1	1
18	1k	CHIP RESISTOR - SURFACE MOUNT 1KOHMS ±1% 1/4W 1206 ROHS	R2, R3, R4, R7, R8, R10, R11, R15	8
19	1M	CHIP RESISTOR - SURFACE MOUNT 1MOHMS ±1% 1/4W 1206 ROHS	R5, R9	2
20	10k	10 kOhms ±5% 0.25W, 1/4W Chip Resistor 1206 (3216 Metric) Automotive AEC-Q200 Thick Film	R6, R12, R13, R14, R16, R17, R20, R23	8
21	1 Ohm	1Ω ±5% 0.25W ±400ppm/℃ 1206 Chip Resistor - Surface Mount RoHS	R18	1
22	10 Ohm	CHIP RESISTOR - SURFACE MOUNT 0.1OHMS ±1% 1/4W 1206 ROHS	R19	1
23	0 Ohm	CHIP RESISTOR - SURFACE MOUNT 0OHMS ±1% 1/4W 1206 ROHS	R21, R22	2
24	SW_2Pin_100mil_Switc	2Pin Header	SW1	1
25	PREC003SAAN-RC	Connector Header Through Hole 3 position 0.100" (2.54mm)	SW2, SW4, SW5	3
26	TL3305AF160QG	Tactile Switch SPST-NO Top Actuated Surface Mount	SW3	1
27		Test Point 300 mil centers	TP1, TP2, TP3	3
28	TP_5V	Test Point 300 mil centers	TP4	1
29	10x Probe TP	Test Point 300 mil centers	TP5, TP6, TP7, TP8, TP9, TP10	6
30	ATMEGA328P-ANR	AVR AVR® ATmega Microcontroller IC 8-Bit 20MHz 32KB (16K x 16) FLASH 32-TQFP (7x7)	U1	1
31	CH340G	Transceiver USB 2.0 2Mbps SOP-16_150mil USB ICs RoHS	U2	1
32	TESD5V0V4UCX6_RF(	SOT-26, 5V, 75W, 0.6pF, ESD Protection	U3	1
33	MCP6002T-I/SN	0.6 V/us 2 1MHz 1.8V ~ 6V 100μA General Purpose SOIC-8_150mil Operational Amplifier ROHS	U4	1
34	DAC IC	IC DAC 12BIT V-OUT SOT23-6	U5	1
35	ADS1115IDGSR	16-Bit ADC with Integrated MUX, PGA, Comparator, Oscillator, and Reference 10-VSSOP -40 to 125	U6	1
36	X322512MSB4SI	SMD Crystal Resonators 12MHz ±10ppm SMD-3225_4P RoHS	Y1	1
37	X322516MLB4SI	±10ppm 16000000Hz -40℃~85℃ 贴片无源晶振 60Ω 9pF SMD3225-4P Crystals ROHS	Y2	1

Napkin sketch:

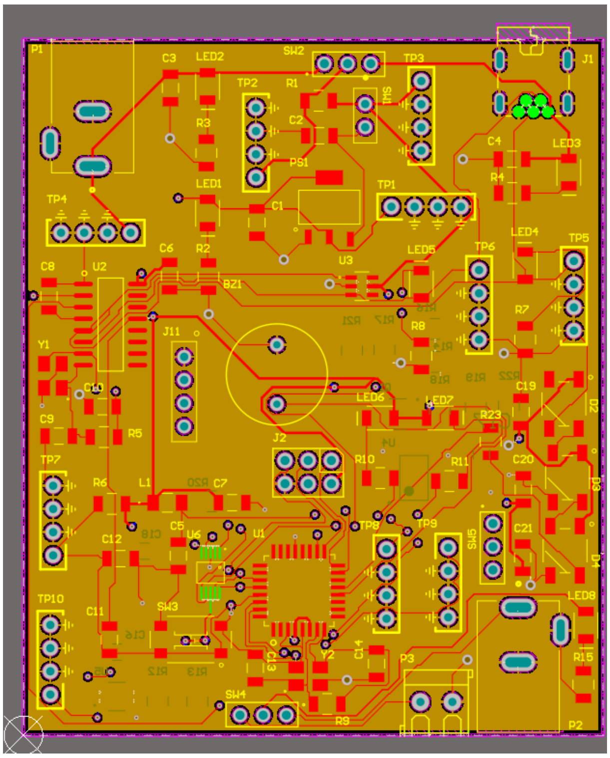


Altium Schematic:





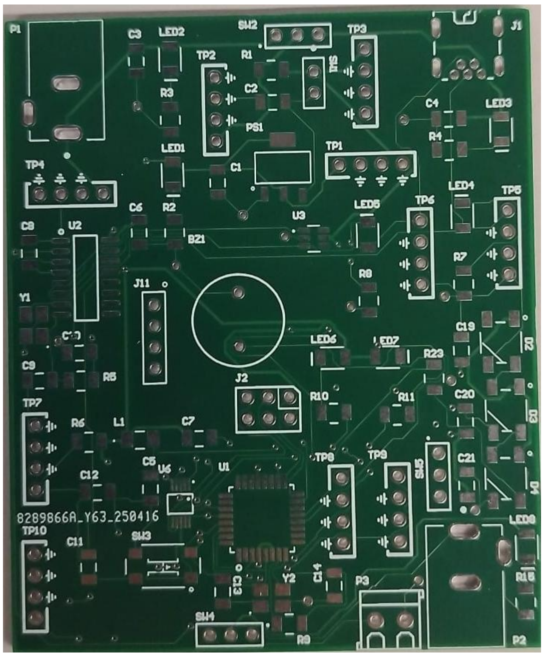
Board Layout:

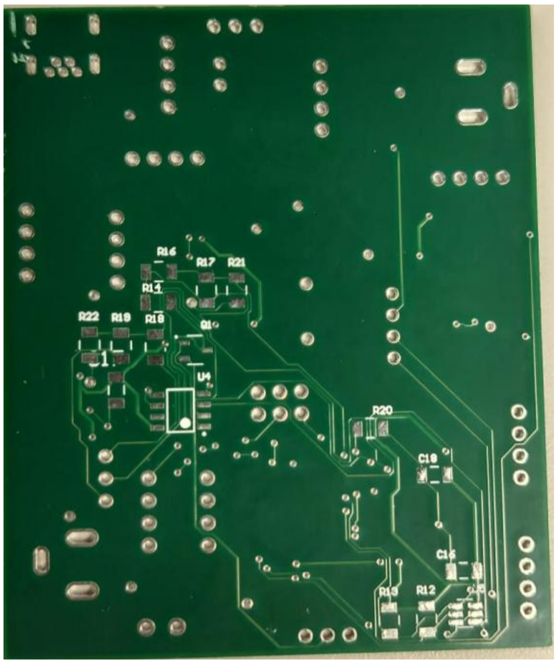


The 4-layer PCB design consists of a top signal layer, two internal ground planes, and a bottom signal layer. The top layer is used for compact signal routing and placement of all surface-mount components. Inner layer 1 serves as the primary ground plane, ensuring low-impedance return paths for top-layer signals and reducing EMI. Inner layer 2 acts as a secondary ground plane, supporting high-current return paths. The bottom layer is utilized for additional signal routing, power distribution, and interfacing with through-hole connectors and test points.

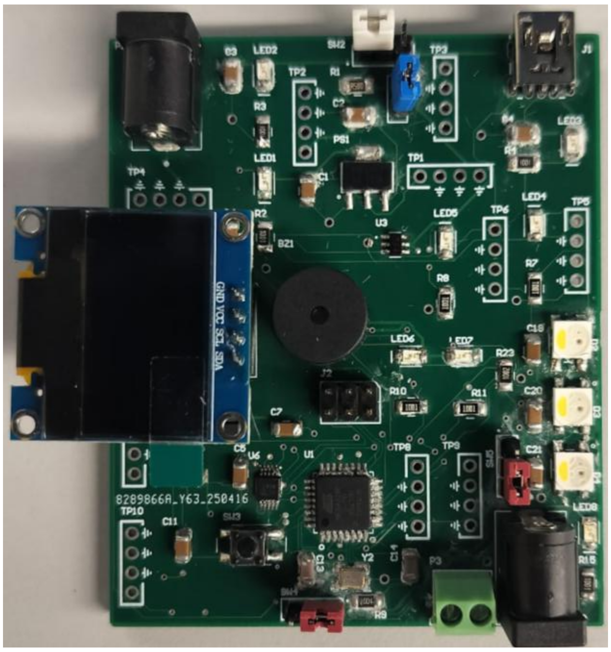
Board Pictures:

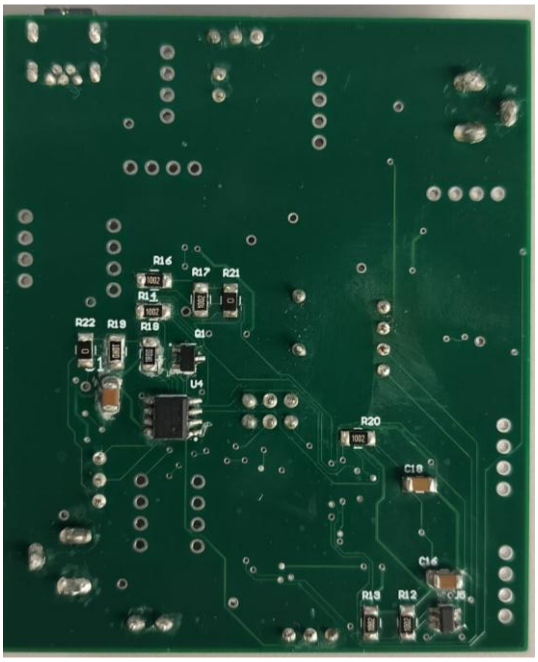
Before assembly:





After assembly:





Plan of Record:

Power

- Power jack for a standard “AC-to-5 V DC” adaptor (primary board supply).
- One more power jack and two-pin screw terminal for the external source whose R_{thev} is being measured.
- On-board slide switch selects either power jack or two pin screw terminal as the local 5 V logic supply.
- 500m ohm current-sense resistor in series with the main rail, plus nearby test pad for inrush-current probing.

Source-Under-Test

- MCP4725 12-bit DAC → MCP6002 rail-to-rail op-amp → MOSFET forms a closed-loop electronic load.
- Slide switch selects 1 ohm or 10ohm sense network at the MOSFET source to extend current range.

Measurement

- Ferrite bead on AVCC isolates the ADC/DAC analog rail from digital noise.

Bootloading

- ATmega328-P clocked by a 16 MHz crystal boot-flash accessible through a 6-pin ICSP header and isolation-header jumpers.

- CH340G (12 MHz crystal) provides USB-to-UART bridge on a mini-B connector protected by a USB-line TVS diode.

User Interface

- OLED displays real-time R_{thev}, load current, and rail voltage.
- Three smart LEDs.
- 5 V buzzer

Connectivity & Debug

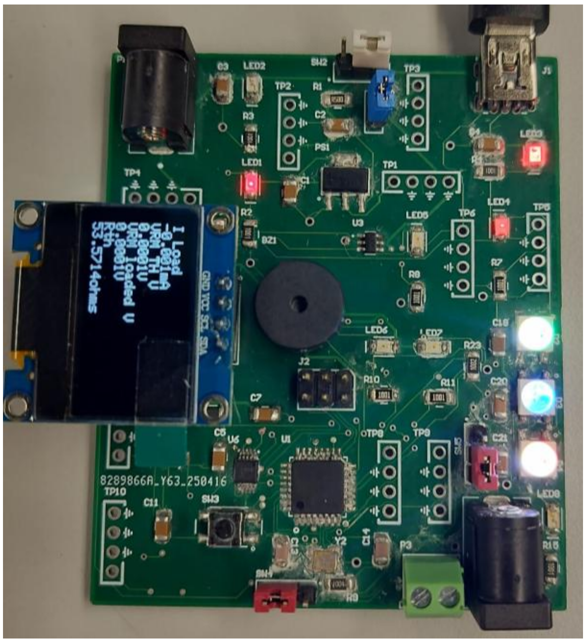
- Labeled test.
- Isolation headers for the ATmega328 I/O and power to support subsystem bring-up.

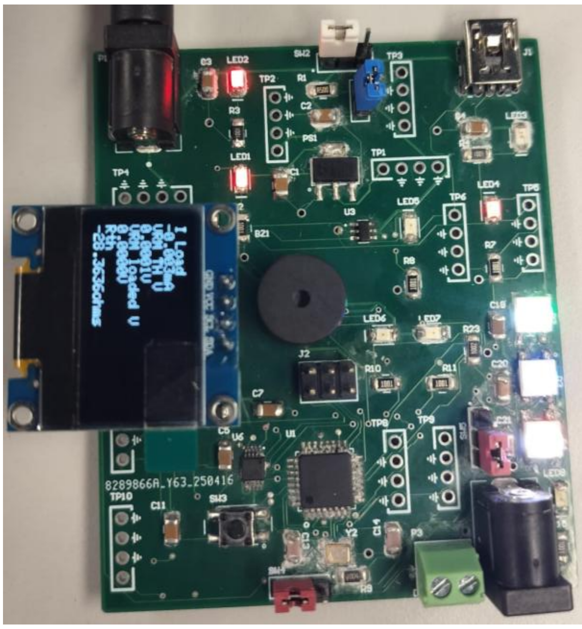
<u>WHAT WORKED:</u>

Board bring up:

1. Power Indicators:

The board was successfully powered up and verified for functionality. The OLED display correctly shows real-time readings of current, voltage, and calculated Thevenin resistance. Multiple indicator LEDs are active, signaling proper power distribution, program execution, and system status. The buzzer is mounted and functional, providing audible feedback during operation. All subsystems—including power input, USB serial interface, Atmega328 bootload, OLED, DAC, ADC, MOSFET load path, and user interface—were confirmed operational during the initial bring-up.



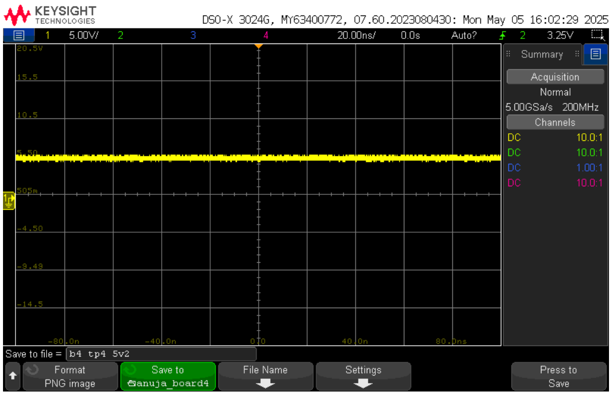


Power on using power USB

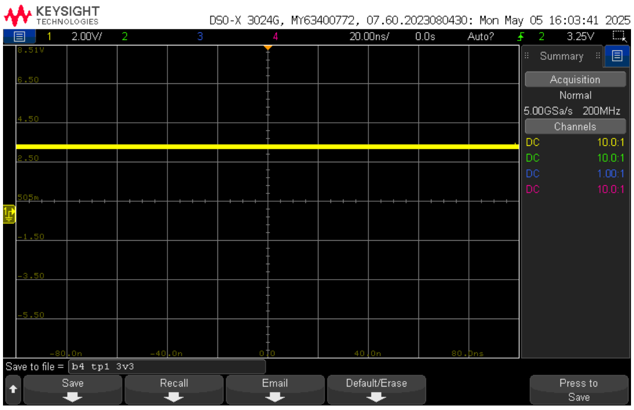
Power on using power jack

2. Power Source Switching:

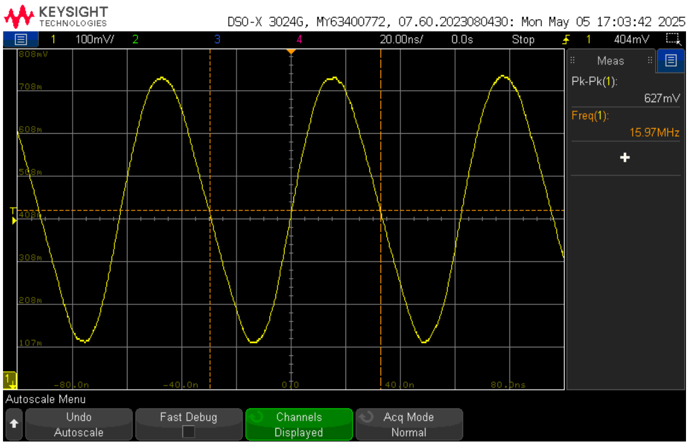
The switch correctly toggles between USB and power (jack), with seamless power transition observed and no voltage dips.



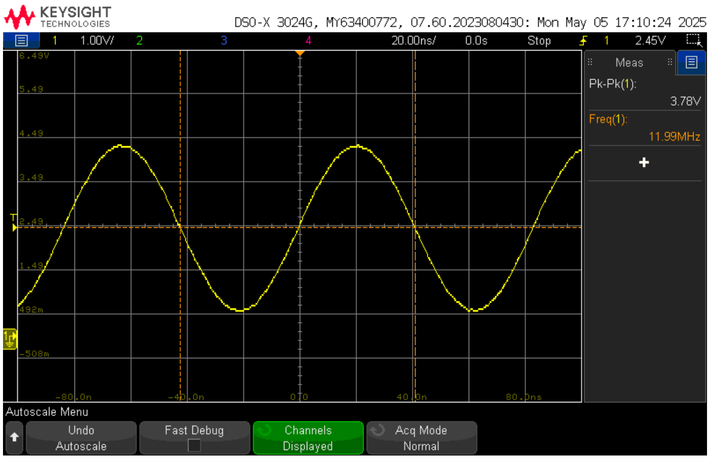
3. 3.3V LDO Output Verification: A stable 3.3V output was observed on the corresponding test point, confirming proper operation of the LDO regulator.



4. Bootloading the ATmega328:
The ATmega328 microcontroller was successfully bootloaded using a commercial Arduino Uno configured as an ISP programmer. Bootloader functionality was confirmed through successful detection in the Arduino IDE and successful sketch uploads. The crystal oscillators were verified for correct operation—16 MHz for the ATmega328 and 12 MHz for the CH340G USB-to-UART converter.

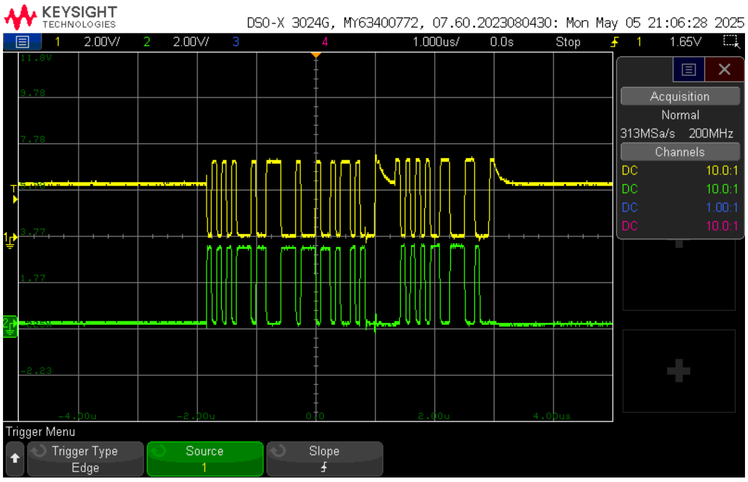


16MHz

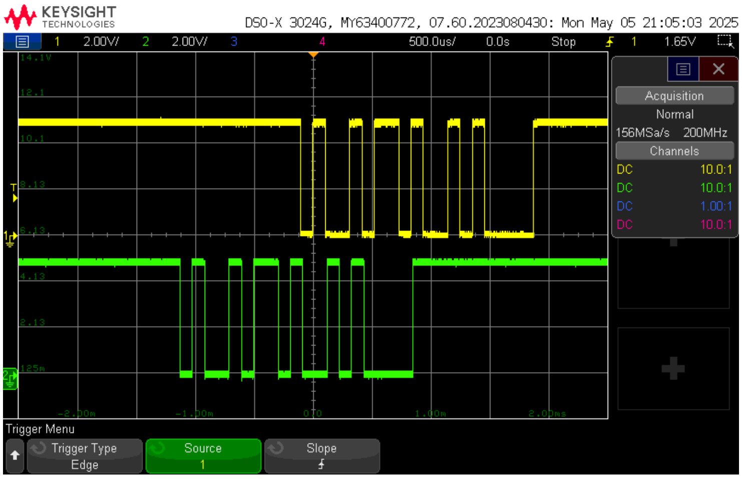


12MHz

5. **Code Flashing and UART Communication:**
Firmware was successfully flashed to the ATmega328 via the CH340G USB-to-UART interface. Oscilloscope captures on USB D+/D– lines and UART RX/TX lines confirmed active data transmission during code upload, verifying proper communication and functionality of the CH340G interface.



D+/D-



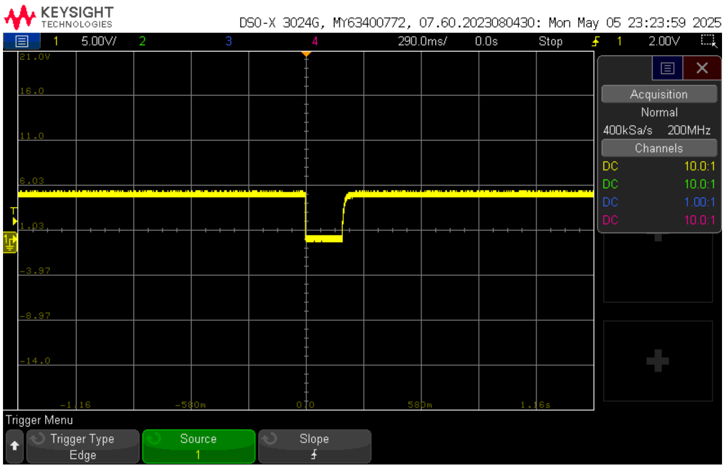
Tx/Rx

6. Flashing Buzzer and Smart LED Test Code:

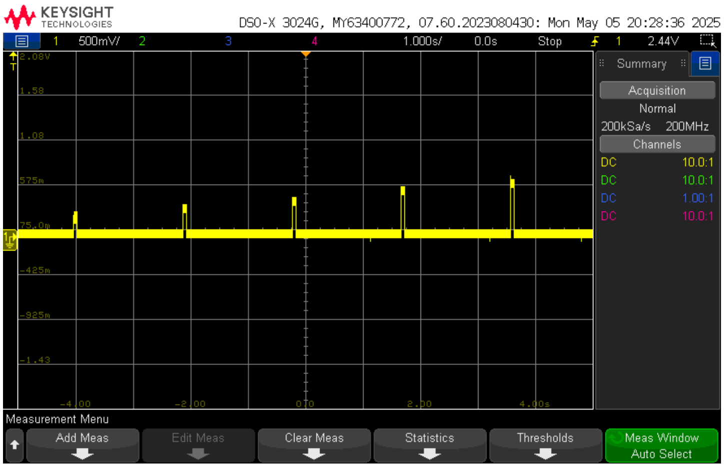
Uploaded a basic test sketch via the Arduino IDE to verify functionality of the buzzer and smart LEDs. The buzzer activated with a short tone and the LEDs cycled through colors. Please find the attached code at the provided link. Check the link for the demo

[demo.mp4]

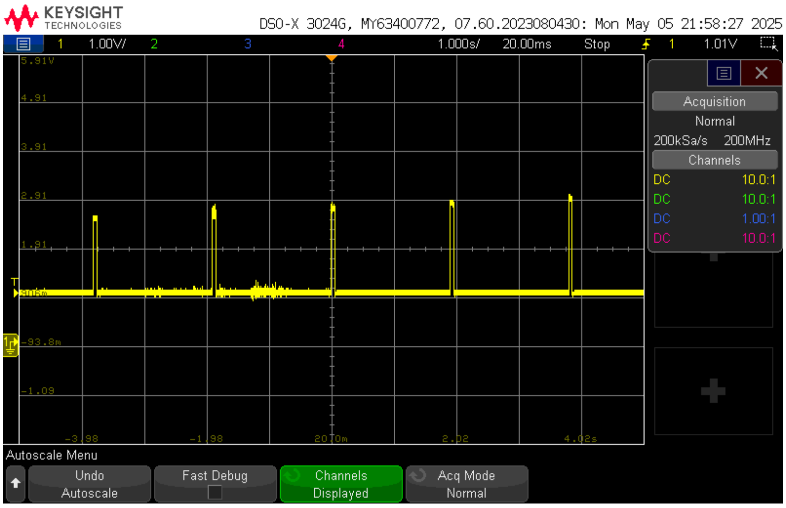
7. Testing reset:



7. DAC output:



8. MOSFET gate

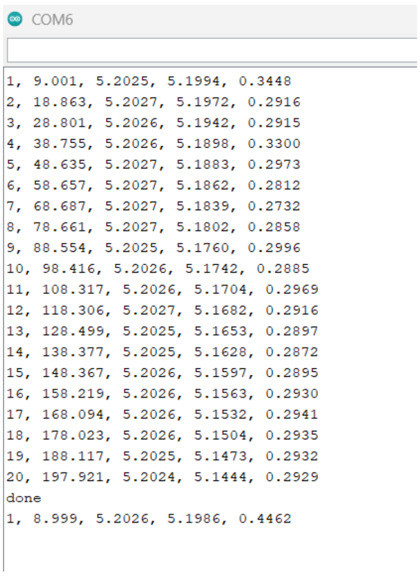


What Did Not Work:

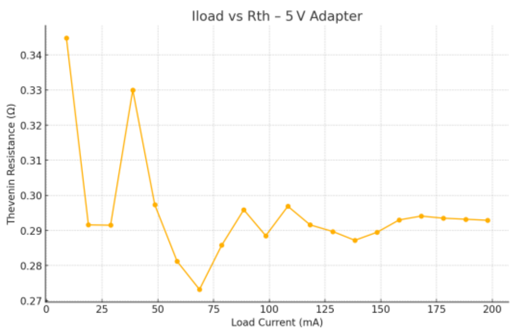
All components and functionalities operated as intended. No hardware or software issues were encountered during bring-up and testing.

Measurement of Output Resistance for Connected Voltage Sources:

1. 5V adapter


COM6

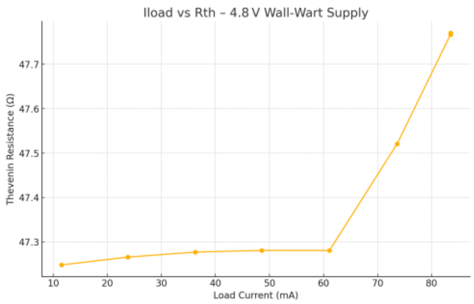
```
1, 9.001, 5.2025, 5.1994, 0.3448
2, 18.863, 5.2027, 5.1972, 0.2916
3, 28.801, 5.2026, 5.1942, 0.2915
4, 38.755, 5.2026, 5.1898, 0.3300
5, 48.635, 5.2027, 5.1883, 0.2973
6, 58.657, 5.2027, 5.1862, 0.2812
7, 68.687, 5.2027, 5.1839, 0.2732
8, 78.661, 5.2027, 5.1802, 0.2858
9, 88.554, 5.2025, 5.1760, 0.2996
10, 98.416, 5.2026, 5.1742, 0.2885
11, 108.317, 5.2026, 5.1704, 0.2969
12, 118.306, 5.2027, 5.1682, 0.2916
13, 128.499, 5.2025, 5.1653, 0.2897
14, 138.377, 5.2025, 5.1628, 0.2872
15, 148.367, 5.2026, 5.1597, 0.2895
16, 158.219, 5.2026, 5.1563, 0.2930
17, 168.094, 5.2026, 5.1532, 0.2941
18, 178.023, 5.2026, 5.1504, 0.2935
19, 188.117, 5.2025, 5.1473, 0.2932
20, 197.921, 5.2024, 5.1444, 0.2929
done
1, 8.999, 5.2026, 5.1986, 0.4462
```



2. Function generator 5V:

COM10

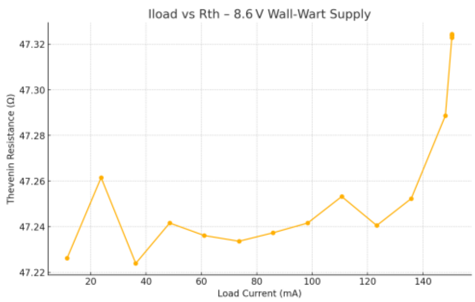
```
1,  11.506,  4.8327,  4.2890,  47.2482
2,  23.785,  4.8327,  3.7084,  47.2657
3,  36.271,  4.8326,  3.1179,  47.2771
4,  48.569,  4.8327,  2.5363,  47.2809
5,  61.093,  4.8327,  1.9441,  47.2807
6,  73.599,  4.8327,  1.3352,  47.5207
7,  83.499,  4.8327,  0.8442,  47.7673
8,  83.498,  4.8327,  0.8441,  47.7689
9,  83.496,  4.8327,  0.8441,  47.7704
10,  83.500,  4.8327,  0.8442,  47.7666
11,  83.499,  4.8326,  0.8442,  47.7665
12,  83.498,  4.8327,  0.8442,  47.7670
13,  83.498,  4.8327,  0.8442,  47.7676
14,  83.498,  4.8327,  0.8442,  47.7678
15,  83.496,  4.8327,  0.8441,  47.7699
16,  83.498,  4.8327,  0.8441,  47.7689
17,  83.498,  4.8327,  0.8441,  47.7691
18,  83.496,  4.8327,  0.8441,  47.7694
19,  83.496,  4.8327,  0.8441,  47.7695
20,  83.495,  4.8327,  0.8441,  47.7700
done
```



4. Function generator 9V:

COM10

```
1,  11.495,  8.6394,  8.0966,  47.2262
2,  23.767,  8.6394,  7.5162,  47.2615
3,  36.270,  8.6395,  6.9267,  47.2239
4,  48.525,  8.6394,  6.3470,  47.2416
5,  61.011,  8.6395,  5.7575,  47.2361
6,  73.554,  8.6395,  5.1652,  47.2336
7,  85.839,  8.6394,  4.5846,  47.2373
8,  98.359,  8.6395,  3.9928,  47.2416
9,  110.640,  8.6394,  3.4113,  47.2532
10,  123.282,  8.6394,  2.8156,  47.2405
11,  135.746,  8.6394,  2.2251,  47.2523
12,  148.081,  8.6394,  1.6369,  47.2887
13,  150.464,  8.6394,  1.5189,  47.3238
14,  150.467,  8.6394,  1.5188,  47.3229
15,  150.465,  8.6394,  1.5189,  47.3238
16,  150.463,  8.6394,  1.5189,  47.3237
17,  150.461,  8.6394,  1.5189,  47.3242
18,  150.463,  8.6394,  1.5188,  47.3244
19,  150.465,  8.6393,  1.5189,  47.3229
20,  150.465,  8.6394,  1.5188,  47.3239
done
```



The instrument droid accurately measured the power adapter near-constant ≈ 47 ohm output resistance and the switch-mode adapter's flat ≈ 0.30 ohm output resistance, demonstrating its ability to distinguish high and low impedance sources.

Conclusion:

The instrument-droid project successfully met all stated objectives: design, fabrication, bring-up, and validation of a standalone 4-layer ATmega328-based board capable of characterizing the Thevenin output resistance of DC sources up to 12 V. Hardware bring-up verified every subsystem—power inputs, USB–UART (CH340G), bootloader access, OLED display, DAC/ADC chain, MOSFET electronic load, smart LEDs, and buzzer—without encountering hardware or software faults.

Key learnings:

GPIO Silkscreen Matters: Unlabeled pins significantly slowed troubleshooting; clear GPIO labels streamline testing and save time.

ICs First: Reflowing the fine-pitch ICs before resistors, capacitors, and jumpers minimizes alignment issues and simplifies later hand-soldering.

Solder Paste Proficiency: Gained hands-on experience soldering components using solder paste, resulting in cleaner and more reliable joints for SMT parts.